

Phishing Email Investigation (TryHackMe)

Overview

This lab was completed in my home SOC lab using a virtual machine. The objective was to investigate a suspicious email, identify phishing indicators, and perform technical analysis using multiple security tools.

Although the TryHackMe room only required one analysis method, I expanded the investigation by using several industry-standard tools to simulate a real SOC Level 1 analyst workflow.

Environment

Platform: TryHackMe

Operating System: Linux Virtual Machine

Role: SOC Level 1 Analyst

Lab Type: Email Phishing Investigation

Tools Used

VirusTotal

DMARCian

MXToolbox

WHOIS Lookup

SHA256 Hashing

Linux Terminal

Email Header Analysis

Investigation Process

1. Email Header Analysis

I first examined the email headers to identify important artifacts including:

Sender email address

Return-Path

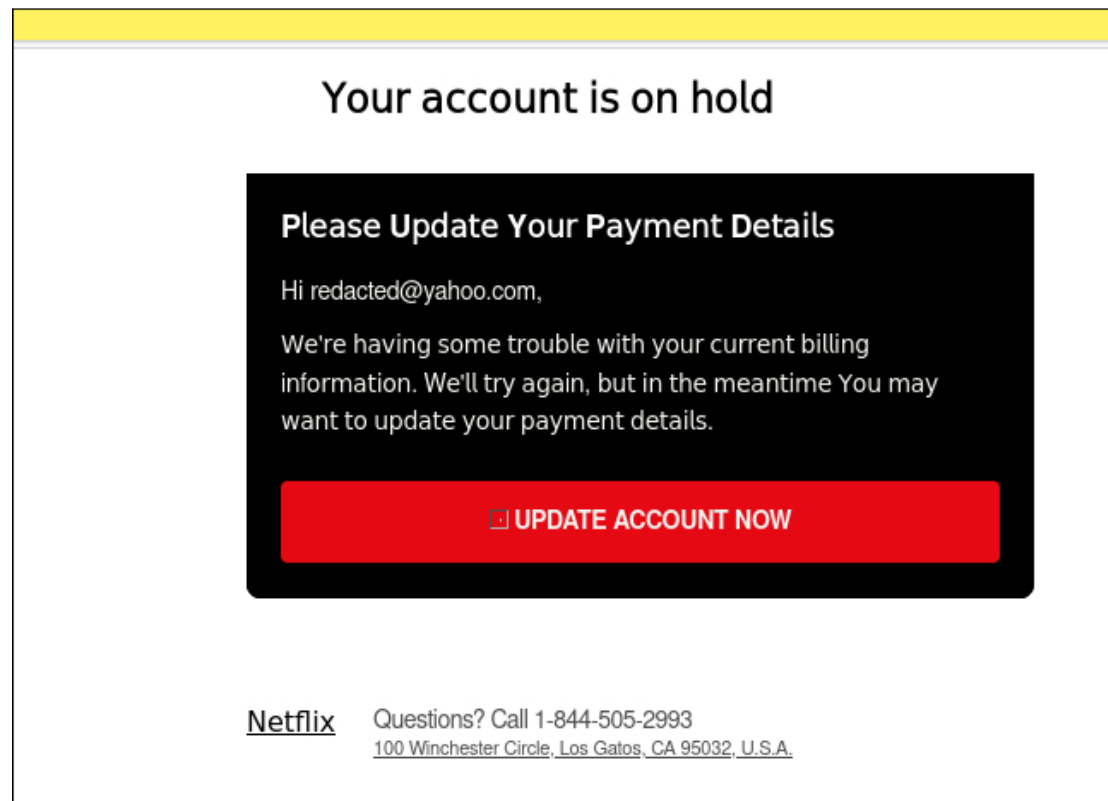
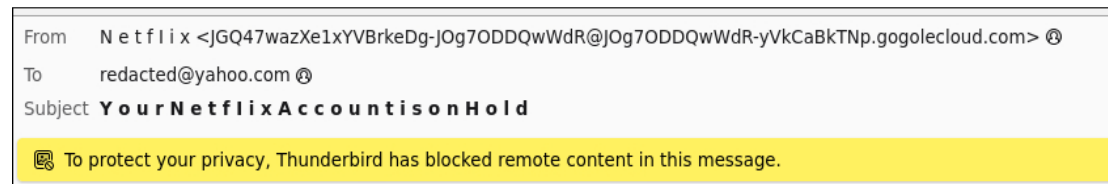
Originating IP address

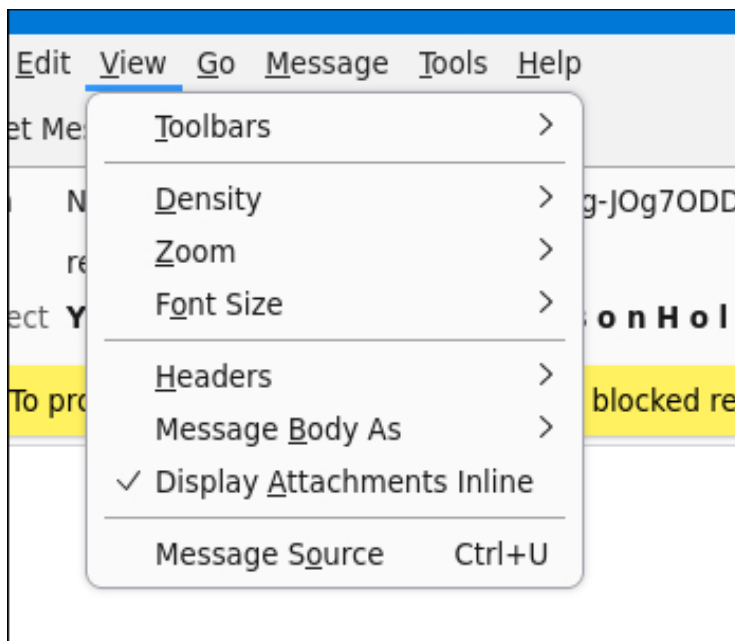
Recipient address

Subject line

Date and time

During this stage, I looked for spoofing attempts and inconsistencies between the sender information and the originating server.





```
Source of: file:///home/ubuntu/Desktop/Phish3Case1.eml?type=appli
File Edit View Help

Received: from 10.197.37.234
  by atlas105.free.mail.bf1.yahoo.com with HTTPS; Wed, 7 Jul 2021 02:14:46 +0000
Return-Path: <postmaster@etekno.xyz>
X-Originating-Ip: [209.85.167.226]
Received-SPF: none (domain of etekno.xyz does not designate permitted sender hosts)
Authentication-Results: atlas105.free.mail.bf1.yahoo.com;
  dkim=unknown;
  spf=none smtp.mailfrom=etekno.xyz;
  dmarc=unknown header.from=J0g70DDQwWdR-yVkJCaBkTnp.gogolecloud.com;
X-Apparently-To: redacted@yahoo.com; Wed, 7 Jul 2021 02:14:47 +0000
X-YMailISG: soSDWIIWLDuoG8NhyrHXR7QH4K4775ng5KvMq.0Ur7XVVwU
Be3WPYvUM0n0esjRJB1FL8aYJlqdiZ9E90kXzwdWdTOYQu60W8AYAEh2
wVHhZe98VYCJmHeR0pIec7.YYXLSVz9FiWUea09lCEmQ0LmI2ELSeQMPxQyi
2qZtWwPhHVubjxbMIRQh3teEA3AlGt7wRZ_j_HwdkS1nnSaFBsnrKZBFwOWl
AmDq.IFtpSufP_Qj3jDDUR3c5xSE9ZjghaA0Udfe4HQ9H_Dc47XAqD.MXAPd
Vy5UXg9I0N8lSmN4d0P72lAIQ1YkMR04c9LQ3LeP9flsqycImfT4M_E0.ChI
AfCaG1SKxytYf0GMVVjSNq..sCJS0bLHY3ZHD8.ddKlJe34v54EABsTv9g8XI
aFatYYdf0MHgWeJ3owQNYH5tT10vzXS33AXecgKPLY0XhACgm0XhNXH5ot6b
YfBvWl_ViqX5nEE13Igt1KQ2Zvsih6dqNPfdK5NHx990dlz82V_Yn2scnqvE
1XnxggAq7pEEgP2N768_DQc_ES7_UpSiTTmkXHMx5T3Eib4dvDhv.YKXu71
GDJFaRc600UTtHWE44FxdVynvKhs_Z2avfKM_m2LFtlonkTeScmZRBefstPC
CkXLPQoWsXVbHbFjPEZXpU4UA9L1Cs_IlnHa7GbIcgx4.Mix2lsMTw.ihbd
68qMFXSUXXiQU0kut7Mw5qJZvYpq8SYaoi0tYyqBiHqQeATq0Mj9sfxPw5o
uRmyXBx6PHNbpwrfjnoxpr_gMKPzyS5dh07pCD10VDWkRNBZWgCWcPSeoq.f
LUK5js1TGLW7e28G4nKHfTUl0yt4fnyM8FbEW58hZNxBSY77wyAhHH1i5HI9
XoiPc3nb_Ma50sVUBti6MNH2ilqJehZ2AACN4zZVFBDT9hdPjqt36A2gDlu2
pAw7uhRduPd553t51qVBvKYTpCDaGzFvWiBVXu4v3ZYg_ykwbtSxLPZ39j
iQXKjjtRPQuaLzbMXJHlenJ4Cfqt05Adm67ukSrflyPQIYplN3a4KfYsbn33
SOMhkpr5tuQYx0qMUKzZevD83XWhbPZ7SiQpjwM7H0gvA--
Received: from 209.85.167.226 (FHL0 mail-oil-f226.google.com)
```

Header Artifacts

Sender email address:

JGQ47wazXe1xYVBrkeDg-J0g70DDQwWdR@J0g70DDQwWdR-yVkJCaBkTnp.gogolecloud.com

Sender IP address: **X-Originating-Ip: [209.85.167.226] AND Return-Path: <postmaster@etekno.xyz>**

Email subject line: **it calls to action Your account is on hold**

Recipient email address: **redacted@yahoo.com**

Reply-To email address: **No**

Date and time: **7 Jul 2021 02:14:46**

Body Analy2. URL Investigation

The email contained a hyperlink prompting the user to update their account.

I submitted the URL to VirusTotal to determine whether it had been detected as malicious by security vendors.sis

URL report for 2026-06-01 08:59:51 UTC. This URL has been scanned 1 time.

<https://t.co/yuxfZm8KPg?amp=1> → <https://bit.ly/2TiRpyj>

t.co 162.159.140.229

text/html

external-resources

trackers

third-party-cookies

Security vendors' analysis

Webroot



Malicious

Categories ⓘ		
alphaMountain.ai		URL Redirect (alphaMountain.ai)
BitDefender		computersandsoftware
Xcitium Verdict Cloud		social networking
Sophos		information technology
Webroot		Phishing and Other Frauds
Forcepoint ThreatSeeker		social web - twitter

4. Technical Defences

I reviewed common enterprise email security controls, including:

Secure Email Gateways (SEG)

Email Filtering

Link Rewriting

Attachment Sandboxing

User Awareness Training

Phishing Reporting

Understanding these controls is essential for SOC analysts when investigating phishing campaigns.

Skills Demonstrated

Email Header Analysis

IOC Identification

URL Reputation Analysis

Domain Authentication

Threat Intelligence

Linux Command Line

SOC Investigation Methodology

Conclusion

This exercise strengthened my understanding of phishing investigations and the workflow followed by

SOC analysts. By combining multiple analysis techniques and security tools, I was able to identify indicators of compromise and determine that the email displayed characteristics consistent with a phishing attack.